

SenseSearch: Empowering Vision-Language Models with High-Resolution Agentic Search-Reasoning via Reinforcement Learning

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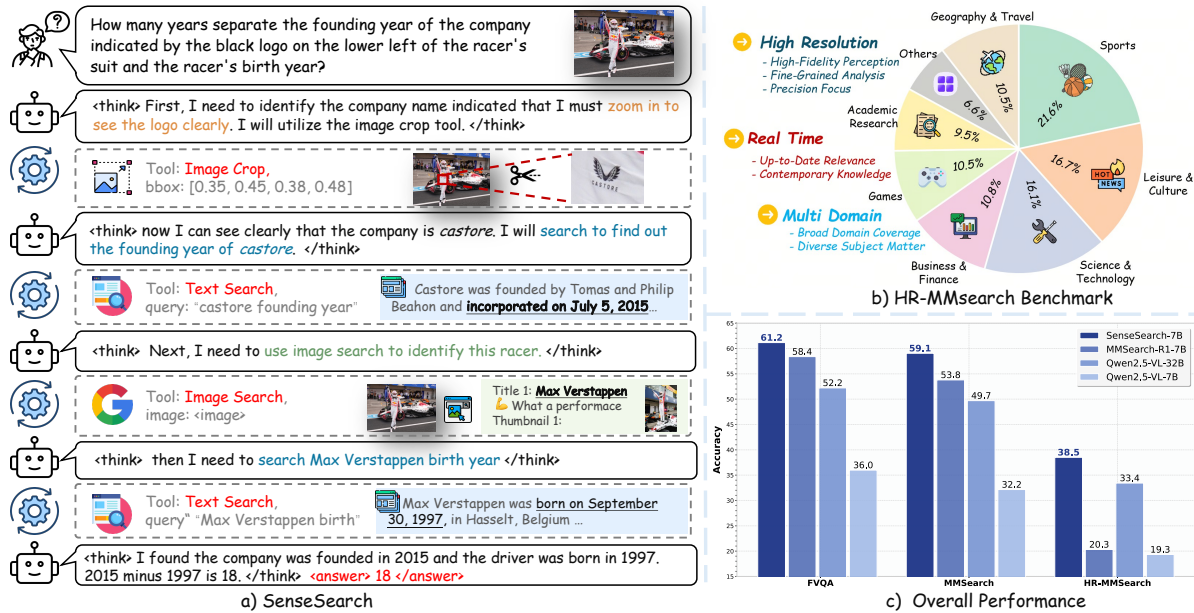


Figure 1. Overview of SenseSearch. (a) SenseSearch tackles the challenging visual task by leveraging an integrated suite of text search, image search, and image crop tools within the reasoning process. (b) Our proposed HR-MMSearch benchmark characterized by the high-resolution images, knowledge-intensive question and diverse scenes. (c) Comparison of VLMs on the search-oriented benchmarks.

Abstract

Vision-Language Models (VLMs) are limited by static knowledge and insufficient fine-grained visual analysis, hindering their performance on knowledge-intensive and visually complex tasks. While recent research has explored VLMs that employ external tools like search or cropping to enhance model performance, they typically employ tools in isolation and lack the ability to coordinate multiple tools effectively. To address this gap, we propose SenseSearch, the first agentic VLM for search-reasoning that supports adaptive multi-tool coordination via reinforcement learning (RL). Specifically, SenseSearch dynamically integrates the image search, text search, and image crop tools to tackle fine-grained and knowledge-intensive visual understanding challenges. We first con-

struct a high-quality cold-start dataset to instill basic tool-usage behaviors. In the subsequent RL stage, we introduce Batch-Normalized Group Sequence Policy Optimization (BN-GSPO) algorithm to enhance the tool invocation and reasoning ability. To comprehensively evaluate the agentic VLMs on complex visual tasks, we introduce the HR-MMSearch benchmark, the first search-oriented benchmark composed of high-resolution images with knowledge-intensive and search-driven questions. Experiments demonstrate that SenseSearch achieves state-of-the-art performance on open-source search and fine-grained image understanding benchmarks, outperforming baselines by 19.18% on HR-MMSearch. SenseSearch provides a promising path toward agentic VLMs with effective and robust tool invocation capabilities. All code and data are released at <https://github.com/OpenSenseNova/SenseNova-MARS>.

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1. Introduction

Vision-Language Models (VLMs) have significantly advanced the development of artificial general intelligence with impressive capabilities across a wide range of visual understanding tasks [1, 4, 7, 15, 41]. However, VLMs are hindered by static knowledge bases for knowledge-intensive tasks, as well as limited fine-grained analysis of high-resolution images. This is particularly problematic in open-world settings, where the prevalence of long-tail knowledge and real-time information creates significant challenges for models reliant solely on a static training corpus [2, 17, 19, 38]. To overcome this limitation, a promising approach is to equip models with the ability to use external tools, such as search engines, for accessing up-to-date information [24, 42]. Effective tool usage further requires strong reasoning skills, enabling model to proactively reason, plan, and interact with tools to solve complex tasks. Therefore, to solve real-world image understanding challenges as shown in Fig. 1(a), it is critical to develop agentic models with robust tool invocation and reasoning capabilities.

The potential of end-to-end reinforcement learning (RL) with Group Relative Policy Optimization (GRPO) to enhance the reasoning capabilities has been highlighted by models such as DeepSeek-R1 [13], establishing RL as the mainstream approach for developing reasoning models. Existing search-based agents [8, 18, 24] have utilized end-to-end RL to enhance model performance through multi-turn reasoning with text search tools, demonstrating substantial performance gains over Retrieval-Augmented Generation (RAG) methods. MMSearch-R1 [42] further integrates image search with text search tools, enabling VLM to interact with the real-world environment to tackle knowledge-intensive and info-seeking visual tasks. However, it remains inadequate for high-resolution perception tasks as shown in Fig. 1(a), which requires a level of detailed visual reasoning and fine-grained analysis that cannot be achieved with image and text search tools alone. Recent advances such as OpenAI-o3 [30] introduced the “*Thinking with images*” paradigm, which interleaves image and text reasoning by employing the image crop tool to perform fine-grained analysis of complex visual scenes [37]. Subsequent work, including Pixel Reasoner [35] and DeepEyes [52], further demonstrates that pixel-space visual reasoning capabilities can be incentivized by RL, enabling VLMs to learn how to interact with image manipulation tools and proactively gather necessary visual information. These approaches are typically restricted to either search-based or image manipulation tools, which hinders their ability to handle dynamic real-world visual tasks requiring both. This limitation necessitates the creation of search-oriented agentic VLMs that can leverage tools like image crop to handle complex visual scenes, driven by end-to-end RL.

To this end, we propose SenseSearch, an agentic VLM

that leverages RL to integrate image search, text search, and image crop tools into a dynamic, multi-turn reasoning process. Specifically, SenseSearch adaptively interacts with the tool sets through reasoning and planning, learning when and how to invoke the search and image crop tools during the iterative reasoning process. We systematically investigate how to build this agentic VLM based on Qwen2.5-VL [1], focusing on three key aspects: training data construction, a two-stage training pipeline including the cold-start stage and RL, and the design of RL algorithm. We first curate the high-quality cold-start and RL training data through a synthesis pipeline with rigorous quality verification. The dataset spans diverse visual tasks, including knowledge-intensive, high-resolution perception, and search-oriented tasks, which necessitate the use of the search tool, image crop tool, or a combination of both. During the initial cold-start supervised fine-tuning (SFT) stage, the model learns basic tool-usage patterns from a minimal dataset of approximately 3,000 samples. This compact dataset is crucial as it establishes a foundation for the subsequent RL stage, particularly in learning to use the tools effectively. To advance the agentic VLM’s capabilities in multi-tool collaboration and reasoning during the RL stage, we propose Batch-Normalized Group Sequence Policy Optimization (BN-GSPO), an extension of the standard GSPO algorithm. Experiments indicate that BN-GSPO can improve training stability for multi-tool rollout trajectories and yield significant performance gains.

To further evaluate visual search agents for complex scenes, we introduce HR-MMSearch, the first search-oriented benchmark composed of high-resolution images with knowledge-intensive and search-driven questions. SenseSearch demonstrates superior performance on HR-MMSearch, far beyond the search-only or crop-only models, and is comparable with proprietary models like Gemini-2.5-Flash [7]. Furthermore, SenseSearch achieves state-of-the-art (SOTA) results across diverse open-source search-oriented and perception benchmarks. It outperforms MMSearch-R1 [42] on search-oriented benchmarks and attains a leading score of 59.06 in MMSearch [17]. For the high-resolution perception benchmark, SenseSearch achieves 73.6 in HRBench-4K [40], exceeding specialized models like DeepEyes [52] and Pixel-Reasoner [35].

To sum up, our contributions are threefold: (1) We propose SenseSearch, the first end-to-end agentic high-resolution vision-language model developed by RL, with the capability of performing image search, text search, and image crop tools within the reasoning process. (2) We introduce HR-MMSearch, the first benchmark specifically designed for high-resolution, search-oriented, and knowledge-intensive tasks, enabling comprehensive evaluation of agentic reasoning and multi-tool invocation capabilities in VLMs. (3) We propose BN-GSPO, a stable and

high-performing RL algorithm for training agentic VLMs in tool invocation. Extensive experiments on diverse open-source benchmarks with SOTA performance demonstrate that SenseSearch exhibits strong reasoning and robust tool invocation, validating the effectiveness of our approach.

2. Related Works

2.1. Search-based Agentic VLMs

To mitigate the restriction of static knowledge bases, large language models have evolved from static retrieval mechanisms to dynamic, tool-augmented reasoning. Early RAG systems such as Dense Passage Retrieval [20] and Retrieval-Augmented Generation [22] rely on static document corpora for text grounding. Later, search-augmented systems such as WebGPT [28], Toolformer [31], and SAIL [27] allow models to access live web information and reason over up-to-date content. In parallel, multimodal models including REVEAL [14], RagVL [5], and VisRAG [47] show that combining textual and visual retrieval improves knowledge-intensive understanding. However, these RAG-based methods follow the fixed workflow and often result in excessive retrieval, which is suboptimal in practice. Recently, text-based agentic search models such as Search R1 [18] and Search o1 [24] integrate search tools into the chain of thought to enhance performance on knowledge-intensive tasks. In the multimodal domain, agentic search models such as MMSearch-R1 [42] train VLMs to trigger image and text searches dynamically, while WebWatcher [12] extends this approach through synthetic trajectories that improve generalization. Existing agentic VLM research has mainly focused on holistic image understanding that gathers broad contextual information while overlooking the analysis of fine-grained image regions. This limitation reduces their ability to answer region-specific questions and to perform tasks that require precise visual grounding.

2.2. Thinking with Images

The “*Thinking with images*” paradigm, introduced by OpenAI-o3 [30], has spurred the development of agentic visual reasoning VLMs that can interleave image and text reasoning with iterative visual analysis [36, 37]. DeepEyes [52] provides an open-source implementation, demonstrating that end-to-end RL can incentivize models to adopt this behavior, significantly improving performance on fine-grained visual tasks. However, subsequent works reveal that pure RL training is insufficient for complex, multi-turn interaction [21, 35, 49, 53]. Pixel Reasoner [35] identifies a critical “learning trap” where models bypass nascent visual tools. To address this, it proposes a two-phase approach: a cold-start phase to first establish foundational tool use, followed by a curiosity-driven RL phase to incentivize pixel space exploration. Similarly, Mini-o3 [21] observes that

pure RL cannot generate the deep trajectories for hard visual searches and therefore adopted a two-stage training method to activate multi-turn capabilities. Despite these advances, current methods remain fundamentally knowledge-limited, as their toolsets focus solely on perceptual image operations and are inherently ill-equipped for tasks that demand open-web access or external knowledge.

2.3. Multi-tool Agentic VLMs

Recent advancements in VLMs [1, 15, 41, 44], particularly the integration of RL paradigms, have catalyzed the rapid evolution of sophisticated agentic systems [9, 16, 39]. These models demonstrate impressive capabilities in complex problem-solving through hierarchical planning, reasoning, and the strategic invocation of diverse external tools (e.g., image processing, code execution, web search). Notably, recent studies [10, 25, 34] have highlighted the crucial role of RL in enabling agents to develop adaptive multi-tool utilization strategies, optimizing both tool-calling policies and reasoning trajectories through interaction with the environment and feedback from execution outcomes. VisualARFT [26] introduces a framework that empowers agents to perform multi-tool collaboration by integrating web information retrieval with autonomous code execution for visual reasoning. Similarly, DeepMMSearch-R1 [29] proposes a VLM-based agent equipped with advanced tool integration for web search, orchestrating textual queries alongside the image search mechanism that leverages intermediate cropping to concentrate on relevant visual entities. However, existing methods struggle to effectively coordinate the acquisition of critical visual information with external knowledge retrieval, limiting their capability for collaborative tool invocation and coherent reasoning.

3. Method

In this section, we investigate how to build an agentic search-reasoning model with fine-grained visual analysis capabilities for complex search-oriented and knowledge-intensive tasks. Our investigation focuses on the two-stage training strategy, data construction, and benchmark evaluation. We first present the formal task formulation in Sec. 3.1. We then describe the two-stage training approach in Sec. 3.2, which consists of cold-start SFT and RL. The data collection pipeline for the SFT and RL stages is detailed in Sec. 3.3. Finally, in Sec. 3.4, we introduce HR-MMSearch, our newly constructed benchmark designed to rigorously evaluate the fine-grained visual analysis and search-reasoning capabilities of VLMs.

3.1. Problem Formulation

We build upon the MMSearch-R1 [42] problem setting for interactive and search-augmented multimodal reasoning in a real web environment, and extend it in two main ways.

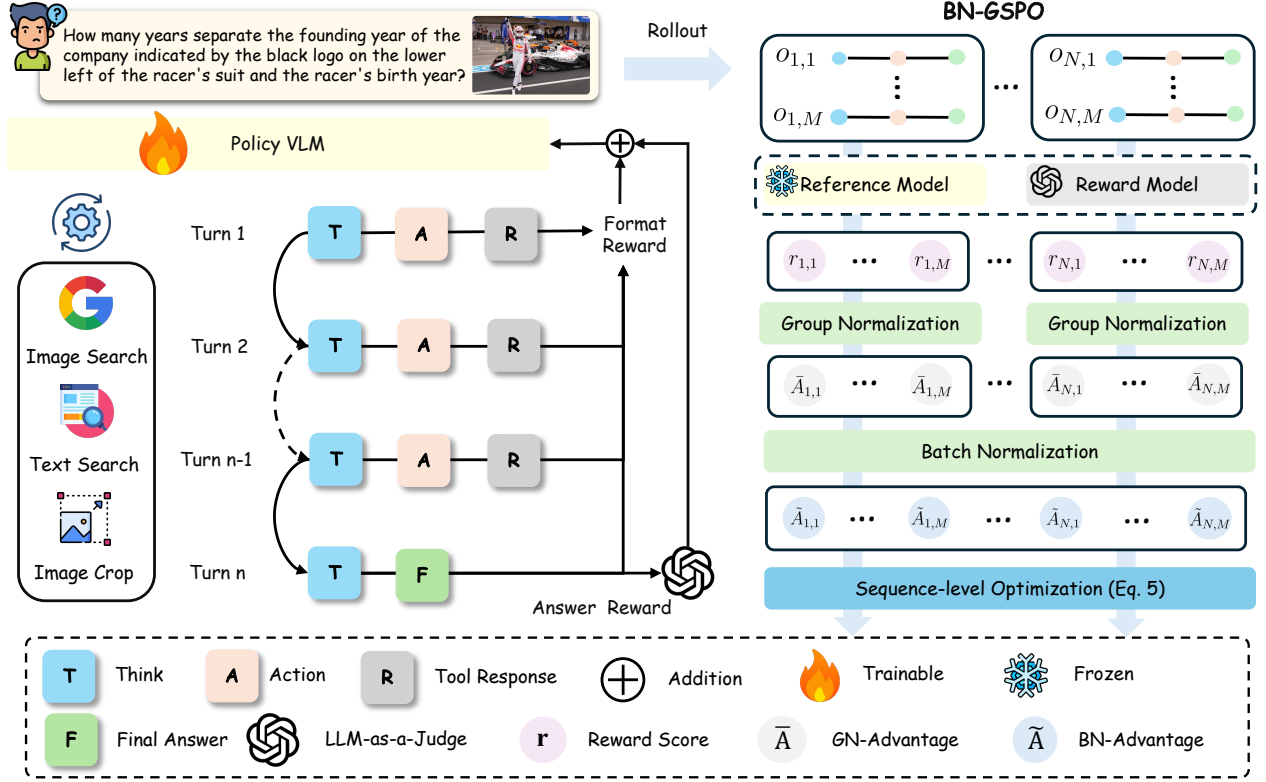


Figure 2. The illustration of SenseSearch RL training pipeline. SenseSearch adaptively invokes the image search, text search and image crop tools in the multi-turn reasoning process to obtain the final answer. The policy VLM is optimized by the BN-GSPO algorithm, driven by the format reward and answer reward.

First, we expand the available toolset beyond text and image search by adding the image crop tool, which allows the agent to zoom into specific regions of an image. Second, we focus on multimodal search tasks that involve high-resolution images, where important information can only be revealed through selective zooming. The left panel of Fig. 2 presents an overview of this setting. It illustrates how the agent interacts with the environment through reasoning and tool use, and how the task, observation space, and action space are structured within this process.

Task and Objective. The task begins with a natural language query q and an initial image I_0 . At each turn, the agent may invoke a tool or produce a final answer when it deems the information sufficient. The process ends once the final answer is given. If no valid answer is produced within T turns, the result is considered incorrect.

Observation Space. At turn t , the model observes the full interaction history $\mathcal{T}_t$, which records all text and image elements in order, including the initial prompt, the query, prior reasoning steps, tool calls, and their outputs. Each tool call yields a compact observation o_t , in text, image, or both, which is then appended to $\mathcal{T}_t$. For instance, calling the image crop tool adds the resulting cropped image. The framework is agnostic to the specific form of these observations,

requiring only that each provides self-contained information for reasoning.

Action Space. At each turn, the model first generates a reasoning step, then selects one of four actions:

1. Perform a text-based web search using *text search*.
2. Execute a reverse image search using *image search*.
3. Zoom into a specific image region using *image crop*.
4. Produce the final answer.

The output, such as search results or a cropped image, is added to $\mathcal{T}_t$, forming an evolving trajectory. Each turn includes a reasoning step and one valid action. If either is missing, the trajectory is invalid.

3.2. Reinforcement Learning with Cold-start

We follow a two-stage training recipe. In the first stage, we perform cold-start SFT to bootstrap the model’s basic ability to learn multi-tool invocations. In the subsequent RL stage, we employ the proposed Batch-normalized GSPO (BN-GSPO) algorithm to further refine the tool invocation and reasoning capabilities.

Cold-start SFT. The cold-start stage involves SFT on a small, meticulously curated dataset $\mathcal{D}_{\text{SFT}}$ of multi-turn interaction trajectories. We formulate the cold-start process

as follows:

$$\mathcal{L}_{\text{SFT}} = - \sum_{(x_i, y_i) \in \mathcal{D}_{\text{SFT}}} \log \pi_{\theta}(y_i | x_i), \quad (1)$$

where $\mathcal{D}_{\text{SFT}}$ is the cold-start dataset, x_i denotes the user query, y_i is the target reasoning trajectory, and π_{θ} represents the model parameterized by θ .

BN-GSPO for RL. We build on GSPO [50] to train agents capable of multimodal search and visual reasoning via external tools. In this setting, each agent output is a sequence that may include text, tool calls, and observations, with supervision provided only at the sequence level through an external reward model. This necessitates sequence-level optimization, for which GSPO is a natural starting point. However, standard GSPO is sensitive to the diverse trajectory structures and reward magnitudes that arise in multimodal and tool-augmented tasks. Different prompts or tool interactions within the same batch can result in varying outcome lengths, reward scales, and difficulty levels, which may bias the learning signal and destabilize training.

To address this issue, we propose BN-GSPO, which applies a two-stage normalization to the advantage estimates. This stabilizes optimization across heterogeneous prompts and preserves consistent learning signals within and across groups. Given a question-answer pair $(x_b, z_b) \sim \mathcal{D}_{RL}$, we sample G responses $\{y_{b,g}\}_{g=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot | x_b)$ and define the length-normalized sequence importance ratio:

$$s_{b,g}(\theta) = \left(\frac{\pi_{\theta}(y_{b,g} | x_b)}{\pi_{\theta_{\text{old}}}(y_{b,g} | x_b)} \right)^{1/|y_{b,g}|} \quad (2)$$

$$= \exp \left(\frac{1}{|y_{b,g}|} \sum_{t=1}^{|y_{b,g}|} \log \frac{\pi_{\theta}(y_{b,g,t} | x_b, y_{b,g,<t})}{\pi_{\theta_{\text{old}}}(y_{b,g,t} | x_b, y_{b,g,<t})} \right).$$

Let $r_{b,g}$ denote the scalar sequence-level reward from the external reward model for $(x_b, y_{b,g})$. We first apply GSPO’s group-level standardization to compute the group-normalized rewards:

$$\bar{A}_{b,g} = \frac{r_{b,g} - \text{mean}(\{r_{b,g'}\}_{g'=1}^G)}{\text{std}(\{r_{b,g'}\}_{g'=1}^G)}, \quad (3)$$

and then normalize these values across the entire optimizer minibatch:

$$\tilde{A}_{b,g} = \frac{\bar{A}_{b,g} - \text{mean}(\{\bar{A}_{b',g'}\}_{b' \in \mathcal{B}, g' \in \mathcal{G}})}{\text{std}(\{\bar{A}_{b',g'}\}_{b' \in \mathcal{B}, g' \in \mathcal{G}})}, \quad (4)$$

where $\mathcal{B}$ denotes the current minibatch and $\mathcal{G} = \{1, \dots, G\}$ is the set of group indices. This second step helps correct inconsistent scales and variances across different prompts within the same batch. With the normalized advantages, we

apply the following clipped sequence-level objective $J(\theta)$:

$$\mathbb{E}_{x_b, \{y_{b,g}\}} \left[\frac{1}{G} \sum_{g=1}^G \min(s_{b,g}(\theta) \tilde{A}_{b,g}, \text{clip}_{\epsilon_{\text{low}}}^{\epsilon_{\text{high}}}(s_{b,g}(\theta)) \tilde{A}_{b,g}) \right] - \beta D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}). \quad (5)$$

This clipped objective stabilizes RL training by preventing excessively large policy updates. Similarly, the KL term helps prevent overfitting by applying a small penalty to deviations from a frozen reference policy π_{ref} .

Reward Modeling. To optimize the learning process in BN-GSPO, we formulate the sequence-level reward as a combination of final answer accuracy and structural format compliance. Specifically, for a trajectory τ , the total sequence-level reward $R(\tau)$ is given by $R(\tau) = R_{\text{acc}}(\tau) + R_{\text{format}}(\tau)$. The accuracy reward $R_{\text{acc}}(\tau)$ measures the semantic agreement between the predicted answer and the ground truth, which is evaluated using an LLM-as-a-judge. The format reward $R_{\text{format}}(\tau)$ guarantees strict compliance with the interaction protocol. Under this protocol, each non-final turn must consist of a reasoning trace and a single tool call. The final turn, in contrast, must contain the reasoning trace and the answer. Additionally, the protocol requires all content to be enclosed within special tags and all tool calls to conform to a specified JSON schema. Full details are provided in the Appendix.

3.3. Training Dataset Collection

For cold-start SFT, we design a structured three-phase data pipeline, as illustrated in Fig. 3. The pipeline begins with data filtering, where we merge the FVQA train set [42], the Pixel-Reasoner warm-start corpus [35], and curated expert-annotated multimodal QA pairs to construct the raw data pool. We then filter this pool using 8 rollouts from Qwen2.5-VL-7B-Instruct, marking a sample as *hard* if the model answers it correctly one time or fewer. For these hard QA samples, we prompt Gemini-2.5-Flash to synthesize complete solution trajectories through tool invocations. Finally, GPT-4o is used to verify format compliance, logical coherence, and answer plausibility. Only the validated results are retained, yielding ~3,000 high-quality samples. For RL, we use FVQA-train [42] together with VisualProbe-train and DeepEyes-4K-train [21], which provides a robust mix of factual question answering and high-resolution visual analysis tasks.

3.4. HR-MMSearch Benchmark Construction

Existing benchmarks, such as FVQA [42] or MMSearch [17], typically utilize standard HD or lower-resolution images and focus on holistic scene understanding, which leaves a critical gap in evaluating an agent’s detailed visual understanding capabilities. Thus, we introduce HR-MMSearch as shown in Fig. 1(b), a benchmark for the

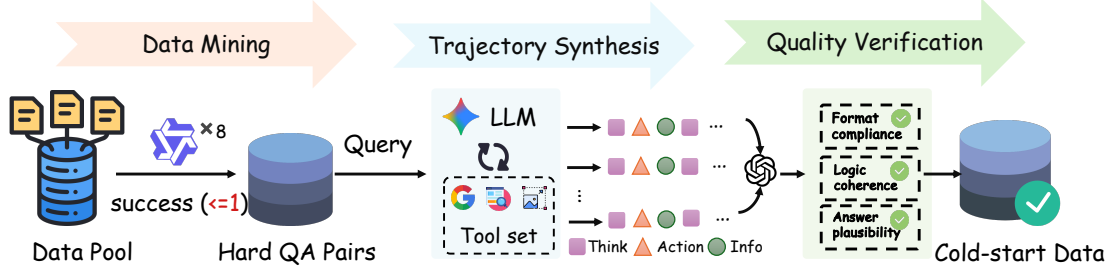


Figure 3. Cold-start data generation pipeline. It consists of data mining, trajectory synthesis and quality verification.

fine-grained perception and search-reasoning capabilities of VLM agents. This dataset consists of 305 4K-resolution images curated from 8 diverse, high-impact domains, covering areas such as Sports, Leisure & Culture, and Science & Technology. To prevent data leakage from the VLMs’ pre-trained knowledge, all images are sourced exclusively from recent 2025 events. For each image, we manually craft knowledge-intensive, search-oriented questions that focus on a key visual subject, such as a small or inconspicuous object or text occupying less than 5% of the image area.

4. Experiments

4.1. Implementation Details

Model and Training. SenseSearch, which is based on Qwen2.5-VL-7B-Instruct [1], is trained using the two-stage pipeline. This process involves SFT followed by RL, which we implement using the LLaMA-Factory [51] and veRL [33] frameworks, respectively. During the SFT stage, we fine-tune only the language model while keeping the vision encoder and multi-modal projector frozen. The training uses a learning rate of 1×10^{-5} for 3 epochs. The SFT stage bootstraps the model’s ability to follow the interaction protocol and utilize the multi-toolset. The subsequent RL stage then teaches the model to use these tools more effectively and efficiently. For RL stage, we use a global batch size of 128, a learning rate of 1×10^{-6} , and a KL coefficient β of 1×10^{-4} . Additionally, to encourage better exploration, we follow DAPO [46] by using the Clip-Higher strategy with $\epsilon_{\text{low}} = 0.2$ and $\epsilon_{\text{high}} = 0.28$. During this RL phase, a single training trajectory allows the agent to interact for a maximum of $T = 10$ turns. This means the agent can iteratively use tools and refine its plan for up to 10 steps before producing a final answer for that trajectory. In each turn, the agent can generate up to 8,192 tokens, with a cumulative limit of 32,768 tokens for the entire trajectory. Detailed training hyperparameters are available in the Appendix.

Training Rewards. To guide the RL process, we employ a two-component reward that jointly promotes answer correctness and format compliance. Each component is a binary score determined by GPT-4o acting as the LLM-as-a-Judge. Specifically, the accuracy reward $R_{\text{acc}}(\tau) = 1.0$ and

format reward $R_{\text{format}}(\tau) = 0.5$ are conferred when their respective criteria are fully satisfied, and are 0.0 otherwise.

Multimodal Tools. SenseSearch can invoke following three multimodal tools using JSON-style arguments:

- *text search*: This tool enables the agent to query the web for textual information. It accepts a single text query as its argument and is powered by the Serper Search API. Following MMSearch-R1 [42], to avoid overwhelming the context length of the model, the top five results are first summarized by Qwen3-32B [45] before being returned to the agent.
- *image search*: This tool performs a reverse image search to retrieve images that are visually similar or contextually related to a given input. It takes an image index as its argument and is powered by the Serper Image Search API. To minimize financial cost and reduce latency during RL training, the top five image search titles and thumbnails for all prompts in the training dataset are pre-fetched and cached in advance.
- *image crop*: This tool allows the agent to crop a previously seen image, allowing focused fine-grained analysis of a specific region of interest. It requires two arguments: normalized coordinates of bounding box [0.0, 1.0] and an index referencing the target image.

Evaluation Benchmarks. To evaluate these capabilities, we test the model on two corresponding categories of benchmarks. For agentic search, we use benchmarks including FVQA-test [42], InfoSeek [3], MMSearch [17], SimpleVQA [6], LiveVQA [11], MAT-Search [26], and our newly constructed HR-MMSearch benchmark. For visual reasoning, we test fine-grained understanding using V* Bench [43], HR-Bench [40] and MME Realworld [48]. Full details on all these datasets are provided in the Appendix.

Baselines. We evaluate our model against several strong baselines. These include proprietary models, such as GPT-4o [15], GPT-4o-mini [15], and Gemini-2.5-Flash [7], as well as open-source agentic models like MMSearch-R1 [42] and DeepMMSearch-R1 [29]. All models are tested under the following three workflow settings:

- *Direct Answer*: The model produces an answer directly without using external tools.

Table 1. Performance on search-oriented benchmarks under Direct Answer, RAG Workflow, and Agentic Model workflows.

Model	Average	MMSearch	HR-MMSearch	FVQA-test	InfoSeek	SimpleVQA	LiveVQA	MAT-Search
<i>Direct Answer</i>								
Qwen2.5-VL-7B-Instruct [1]	27.70	7.60	0.58	26.28	31.95	47.88	19.63	60.00
Qwen2.5-VL-32B-Instruct [1]	32.01	11.70	3.93	30.50	36.65	48.57	21.40	71.33
GPT-4o-mini [15]	33.08	15.79	1.31	36.83	35.95	44.42	24.63	72.66
Gemini-2.5-Flash [7]	40.87	21.64	7.54	43.78	44.10	55.48	31.57	82.00
GPT-4o [15]	42.38	23.39	13.11	48.00	52.90	51.73	28.18	79.33
<i>RAG Workflow</i>								
Qwen2.5-VL-7B-Instruct [1]	50.04	44.44	24.26	55.56	46.10	61.99	42.59	75.33
Qwen2.5-VL-32B-Instruct [1]	58.09	56.14	34.42	63.22	54.60	66.14	50.78	81.33
GPT-4o-mini [15]	58.36	58.48	31.48	64.67	56.3	63.87	51.75	81.33
Gemini-2.5-Flash [7]	62.29	64.91	43.61	65.50	57.75	67.52	54.72	82.00
GPT-4o [15]	63.47	64.33	41.97	69.28	61.45	67.32	57.27	82.67
<i>Agentic Model (zero-shot)</i>								
Qwen2.5-VL-7B-Instruct [1]	35.50	32.16	19.34	36.00	28.80	42.35	22.52	67.33
Qwen2.5-VL-32B-Instruct [1]	53.45	49.71	33.44	52.22	50.10	65.15	42.17	81.33
GPT-4o-mini [15]	45.65	38.60	26.23	50.00	42.35	50.84	31.54	80.00
Gemini-2.5-Flash [7]	58.05	59.06	40.00	61.72	53.70	68.81	47.75	75.33
GPT-4o [15]	60.93	61.40	35.08	67.17	59.55	68.71	48.61	86.00
<i>Agentic Model</i>								
Visual-ARFT [26]	40.13	34.50	24.92	41.72	37.95	42.45	25.40	74.00
DeepMMSearch-R1 [29]	–	–	–	–	47.51	55.87	–	–
MMSearch-R1 [42]	52.49	<u>53.80</u>	20.33	<u>58.40</u>	<u>55.10</u>	57.40	<u>48.40</u>	74.00
SenseSearch-SFT	53.06	<u>53.80</u>	<u>29.80</u>	56.72	52.80	<u>58.63</u>	44.98	<u>74.67</u>
SenseSearch-RL	57.43	59.06	38.52	61.17	55.23	61.20	48.47	78.33

- *RAG Workflow*: Image search results are pre-fetched and provided to the model along with the input question and image. Other tools may still be used.
- *Agentic Model*: The model is provided with all three tools and autonomously decides how to use these tools in the rollout reasoning process.

Details for these workflows are provided in the Appendix.

Evaluation Metrics. For agentic search tasks, the primary metric is Pass@1, assessed by a GPT-4o judge that scores the model’s final answer against ground-truth results. For visual understanding benchmarks, we report Avg@8 Exact Match on V* Bench and HR-Bench, and Pass@1 Exact Match on the large-scale MME-RealWorld benchmark. The evaluation prompt is provided in the Appendix.

4.2. Main Results

Search-oriented Evaluation. As shown in Tab. 1, after the initial cold-start stage, SenseSearch-SFT already matches the performance of the previous SOTA method, MMSearch-R1 [42]. With subsequent RL fine-tuning, SenseSearch-RL establishes a new SOTA among open-source agentic models under 7B parameters, surpassing MMSearch-R1 by an average of 4.94 points. Compared with closed-source propri-

etary models, it outperforms GPT-4o-mini by 11.78 points on average and performs on par with Gemini-2.5-Flash and GPT-4o. These results highlight the efficacy of our unified agentic search and visual reasoning RL framework.

Fine-grained Visual Understanding. As shown in Tab. 2, SenseSearch demonstrates superior fine-grained perception capabilities on high-resolution benchmarks, validating the effectiveness of our approach in detailed, pixel-space analysis. SenseSearch-RL surpasses both specialized tool-based models like Pixel Reasoner [35] and proprietary models like GPT-4o. This strong performance is built upon our two-stage methodology. The initial SenseSearch-SFT model establishes a robust baseline, yielding a substantial +6.6 point gain over the Qwen2.5-VL-7B-Instruct [1], confirming the efficacy of our cold-start dataset. The subsequent RL stage, powered by BN-GSPO, provides the critical refinement, significantly enhancing the SFT model’s performance.

4.3. Ablation Study

Effectiveness of Proposed BN-GSPO. We evaluate the effectiveness of our proposed BN-GSPO by comparing it against the commonly used RL algorithms GRPO and GSPO in Tab. 3. To isolate the influence of SFT, this com-

Table 2. Performance on visual understanding benchmarks.

Model	Size	V* Bench	HR-Bench 4k	HR-Bench 8k	MME RealWorld	Avg.
<i>Direct Answer</i>						
GPT-4o [15]	-	81.2	73.4	70.4	61.0	71.5
LLaVA-OV [23]	7B	75.4	63.0	59.8	57.4	63.9
Qwen2.5-VL [1]	7B	75.3	65.5	62.1	56.8	64.9
Qwen2.5-VL [1]	32B	80.6	69.3	63.6	59.1	68.2
<i>Agentic Model</i>						
SEAL [43]	7B	74.8	-	-	-	-
Pixel-Reasoner [35]	7B	84.3	72.6	66.1	64.4	71.9
DeepEyes [52]	7B	83.3	<u>73.2</u>	69.5	<u>64.1</u>	<u>72.5</u>
SenseSearch-SFT	7B	82.2	72.8	67.3	63.6	71.5
SenseSearch-RL	7B	<u>83.8</u>	73.6	69.8	63.9	72.8

Table 3. Effectiveness of Proposed BN-GSPO.

Method	MMSearch	V* Bench	HR-Bench 4K
GRPO [32]	50.88	67.54	61.38
GSPO [50]	53.80	53.93	44.50
BN-GSPO	56.72	79.05	69.12

Table 4. Impact of different data distribution on RL performance.

Search	Perception	MMSearch	HR-MMSearch	V* Bench
SenseSearch-SFT		53.80	29.80	82.20
✓		54.97	36.80	82.72
	✓	54.09	33.11	85.24
✓	✓	59.06	38.52	83.84

parison uses a pure RL setup where all models are initialized from Qwen2.5-VL-7B-Instruct [1] without a cold-start. As shown, BN-GSPO achieves the best overall performance across all benchmarks and demonstrates stable, balanced improvements. In contrast, while GRPO and GSPO improve on some tasks, they fail to perform well across both search and perception. This indicates that batch normalization effectively mitigates reward scale variance, leading to more consistent and robust multi-tool RL training.

Impact of different data distributions on RL performance. We evaluate the impact of RL data distributions in Tab. 4. Training the model only on specialized data, such as fine-grained perception datasets, makes the agent overspecialize. This boosts V* Bench performance to 85.24 but causes a major drop on search-oriented tasks compared to the SFT baseline. Using our full hybrid dataset, which includes both search and perception tasks, gives the best results by a wide margin on search-oriented metrics. These findings show that hybrid training data is essential. It provides the signals the agent needs to learn a unified, multi-tool policy that avoids overspecialization and balances external knowledge retrieval with fine-grained visual analysis.

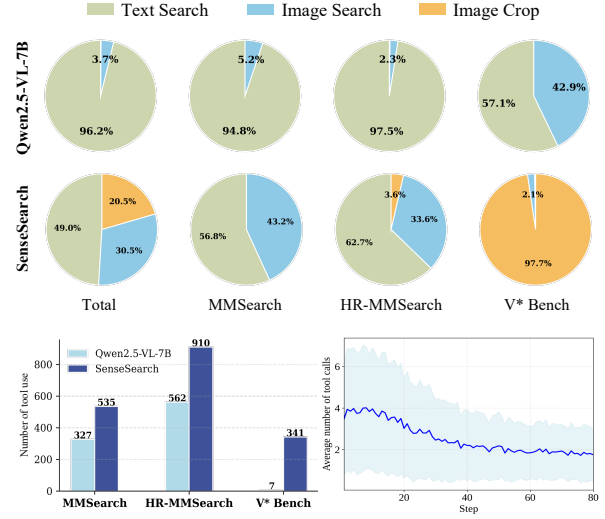


Figure 4. Analysis of tool use behavior. Top: Distribution of tool calls across different benchmarks. Bottom Left: The tool use number in different benchmarks. Bottom Right: Evolution of tool call frequency in the RL training process, indicating that SenseSearch learns more efficient tool invocation strategies.

Analysis of tool use behavior. We evaluate the agent’s tool use behavior in Fig. 4. The base model shows an extreme bias. It relies almost exclusively on text search and ignores the image crop tool, making it ineffective for fine-grained perception. In contrast, SenseSearch shows a strong multi tool ability that adapts well to task demands. SenseSearch identifies the best strategy for each benchmark. It relies almost entirely on image crop for the fine grained perception task in V* Bench. It uses only search tools for the search-oriented MMSearch. It adopts a mixed tool use strategy for the complex SenseSearch. As RL training progresses, the average number of tool calls decreases steadily from ~4 to ~2. This reduction shows that our RL optimization refines the agent’s policy by removing unnecessary actions. It improves the agent’s capability and its operational efficiency.

5. Conclusion

In this work, we propose SenseSearch, a novel search-reasoning agentic VLM that can actively employ the image search, text search and image crop tools within the multi-turn reasoning process for search-driven and fine-grained visual tasks. To empower this capability, we introduce the BN-GSPO algorithm to enhance the reasoning robustness and tool-use proficiency. Furthermore, we construct the HR-MMSearch Benchmark with high-resolution images and knowledge-intensive questions to evaluate the performance of VLMs. Extensive experiments demonstrate that SenseSearch achieves superior performance across diverse benchmarks, showcasing efficient tool invocation and robust reasoning capabilities.

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